

Notes on $c=1$ MQM and entanglement entropy

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ABSTRACT: ...

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1 Review of c=1 MQM

1.1 Continuum theory

From the sigma model to the gauge fixed action. We consider a free scalar coupled to two-dimensional Euclidean gravity, with a cosmological constant and a topological Einstein term on a smooth, compact two dimensional manifold $\mathcal{M}_h$ with genus h and metric g .

$$\mathcal{S}_\sigma^{(h,g,\Phi_0)} = \frac{1}{4\pi} \int_{\mathcal{M}_h} d^2x \sqrt{g} (g^{\mu\nu} \partial_\mu X \partial_\nu X + \Phi_0 \mathcal{R} + 4\pi\mu) . \quad (1.1)$$

Φ_0 is a constant and as we will see later is connected to the constant part of the linear dilaton background. The Gauss-Bonnet theorem states that $\frac{1}{4\pi} \int d^2x \sqrt{g}$ is a topological number known as the Euler characteristic $\chi_h = 2 - 2h$. The partition function is defined through the path-integral over all possible metrics on $\mathcal{M}_h$ modulo Diffeomorphism and the boson X

$$\mathcal{Z} = \sum_{h \geq 0} (g_0)^{-\chi_h} \int \frac{[Dg_{\mu\nu}^{(h)}]}{\text{Vol}(\text{Diffeo})} [DX] e^{-\mathcal{S}_\sigma^{(h,g)}} , \quad (1.2)$$

where we introduced the notation $g_0 = e^{\Phi_0}$. Note that we remove the Φ_0 from $\mathcal{S}^{(\sigma,h,\Phi_0)}$ since we have extracted the topological dependent factor from the action. We can gauge fix the metric to the fiducial metric $g_{\mu\nu} = e^{2b\phi} \hat{g}_{\mu\nu}^{(\tau)}$. The price to pay is the introduction of b, c ghost fields. The measure is not invariant under the previous map. Also, the cosmological constant term yields a non trivial ϕ dependence (because of the cc term, even classically Weyl symmetry is broken). Combined together, they yield the Liouville action

$$\mathcal{S}_L^{(\hat{g}_h)} = \frac{1}{4\pi} \int d^2x \sqrt{\hat{g}_h} \left(\hat{g}_h^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + Q \hat{\mathcal{R}} \phi + 4\pi\mu e^{2b\phi} \right) . \quad (1.3)$$

We emphasize that the term in front of the Ricci scalar term comes from the anomaly of the measure. The remaining integral to perform is an integral over the conformal mode ϕ known as the Liouville field and an integral over the moduli space parametrized by τ introduced previously.

$$M_h = \frac{\text{Metrics on } \mathcal{M}_h}{\text{Diffeo} \times \text{Weyl}} . \quad (1.4)$$

Hence, if modular invariance is satisfied, the partition function finally reads

$$\mathcal{Z} = \sum_{h \geq 0} (g_0)^{-\chi_h} \int d\tau \int [D\phi] [DX] [Db] [Dc] e^{-(\mathcal{S}_\sigma + \mathcal{S}_{b,c} + \mathcal{S}_L)} , \quad (1.5)$$

where V_h is the volume of the moduli space of M_h . Let's focus on the Liouville sector for a moment. We can exploit that ϕ is integrated over to extract the μ dependence of the Liouville partition function. In fact, assuming the measure is invariant under $\phi \rightarrow \phi - \frac{\log(\mu)}{2b}$, one obtains

$$\mathcal{Z}_L^{(h)}(\mu) = \mu^{-\frac{Q\chi_h}{2b}} \mathcal{Z}_L^{(h)}(\mu = 1) . \quad (1.6)$$

Liouville action and Target space interpretation The different terms appearing in the Liouville+matter action have a meaning in terms of Target space fields. Let's recall the general form of the generalized non linear sigma model in presence of a dilaton $\Phi(Y)$, a Tachyon condensate $T(Y)$ and non trivial metric $G_{\mu\nu}(Y)$.

$$\mathcal{S} = \frac{1}{4\pi} \int d^2x \sqrt{\hat{g}} \left(T(Y) + Q\hat{\mathcal{R}}\Phi(Y) + g^{ab}G_{\mu\nu}\partial_a Y^\mu \partial_b Y^\nu \right). \quad (1.7)$$

We see by comparison to the Liouville + matter action that $Y = (X, \Phi)$ and

$$T = \mu e^{2b\phi} \quad , \quad G_{\mu\nu} = \delta_{\mu\nu} \quad , \quad \Phi = \frac{\Phi_0}{Q} + \phi. \quad (1.8)$$

This is important to note that the background (1.8) is solution of the one-loop vanishing of the β functions only for $\mu = 0$. For $\mu \neq 0$ the Liouville interaction should be interpreted as a tachyon-wall background rather than as a naive linear-dilaton solution of the free beta-function equations.

There is also an easy way that the potential plays the role of a Tachyon condensate. Indeed, the vertex operator reads

$$\mathcal{V}_T = e^{ik.Y} := e^{ikX + 2\alpha\phi}. \quad (1.9)$$

The marginality condition fixes $\Delta_X + \Delta_\phi = \Delta_X + \alpha(Q - \alpha) = 1$. Solving the previous condition with $\Delta_X = 0$ and selecting the state satisfying the Seiberg bound one obtains $\alpha = b$. We denote the integrated vertex operator as

$$V_T = \int d^2x \sqrt{\hat{g}} \mathcal{V}_T. \quad (1.10)$$

The Tachyon condensate results in an insertion of the infinite sum of $(\mu V_T)^n$

$$\sum_{n=0}^{\infty} \frac{\mu^n}{n!} \int d^2x_1 \sqrt{\hat{g}} \dots \int d^2x_n \sqrt{\hat{g}} e^{2b\phi(x_1)} \dots e^{2b\phi(x_n)}. \quad (1.11)$$

Remark that we effectively started with a sigma model with one boson and landed on a theory which has now two bosons X, ϕ . We say that from the non-decoupling of the conformal mode, a new dimension has emerged. This describes an inhomogeneous 2D space (ϕ, X) . Such inhomogeneity emerges in the ϕ direction due to the Liouville potential. The string coupling depends on the value of the dilaton and thus on the position in the ϕ direction as the dilaton and the Liouville conformal mode are related by (1.8). An observer sitting on the Liouville wall defined by the condition

$$\mu e^{2b\phi_\star} = 1 \Rightarrow \Phi_\star = \frac{\Phi_0}{Q} + \phi_\star = \Phi_0 - \frac{\log(\mu)}{2b}. \quad (1.12)$$

observes the scattering of states with the string coupling $g_\star = e^{Q\Phi_\star}$. The amplitude of the process should be perturbatively described by a genus expansion in the string coupling at the position of the Liouville wall and this is exactly what the KPZ scaling relation gives (see (1.5), (1.6)).

Genus 0 Lagrangian On the sphere $h = 0$, we choose a representant of the conformal class $\hat{g}_h$ such that the metric is flat on the complex plane $(z, \bar{z})$ except at one point which is taken at infinity. If one writes $ds^2 = 2dzd\bar{z}$, then the Liouville Lagrangian density on the sphere is

$$\begin{aligned} L_{\text{bulk}} &= \frac{1}{4\pi} \left(\partial_z \phi \partial_{\bar{z}} \phi + Q\hat{\mathcal{R}}\phi + \mu e^{2b\phi} \right) \\ &= \frac{1}{4\pi} \left(-\phi \partial_z \partial_{\bar{z}} \phi + Q\hat{\mathcal{R}}\phi + \mu e^{2b\phi} + \text{total derivative} \right). \end{aligned} \quad (1.13)$$

Choosing the representative of the conformal class which is flat on $\mathbb{C}$ trades the bulk curvature coupling for an asymptotic boundary condition at $z = \infty$. Thus the curvature term does not appear

explicitly in the local Lagrangian density on the finite plane, but its effect is retained through the background-charge condition on ϕ at infinity.

$$\phi(z, \bar{z}) \underset{z \rightarrow \infty}{=} -2Q \log(z\bar{z}) + \mathcal{O}(1). \quad (1.14)$$

In fact, $\partial_z \partial_{\bar{z}} \log(z\bar{z}) = 2\pi \delta^2(z, \bar{z})$.

Adding a boundary to the manifold Consider now the possibility of adding a boundary and a non trivial interaction.

$$\mathcal{S}_L^{\text{bound.}} = \frac{1}{4\pi} \int d^2x \sqrt{\hat{g}_h} \left(\hat{g}_h^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + Q \hat{\mathcal{R}} \phi + 4\pi \mu_b e^{2b\phi} \right) + \frac{1}{2\pi} \int_{\partial M} ds \sqrt{\hat{h}} \hat{K} \phi + \mu_b \int_{\partial M} ds \sqrt{\hat{h}} e^{b\phi}. \quad (1.15)$$

h is the physical metric at the boundary and is related to the fiducial metric via $\sqrt{h} = e^{b\phi} \sqrt{\hat{h}}$ and $\hat{K}$ is the extrinsic curvature at the boundary. μ_b is the boundary cosmological constant but should also be understood as the tension/charge of the associated brane. The variational principle is well defined as long as it satisfies Dirichlet boundary conditions at the boundary. It can also satisfies Neumann boundary conditions if

$$n^\mu \partial_\mu \phi + \hat{K} + 2\pi b \mu_b = 0. \quad (1.16)$$

2 c=1 MQM

From the continuum formulation to a discretize model In two dimensions, the Einstein–Hilbert term is topological. Therefore, it is natural to replace the continuum sum over geometries by a sum over discretized random surfaces, such as triangulations, because the discretization can correctly organize the sum over topologies. This is a special feature of two dimensions. Of course, for the continuum limit to be well defined, it should not matter the way we discretize the surface. All the discretized versions should flow to the same continuum limit. This is a crucial feature of the continuum description. The argument showed in what follows should not be taken as a rigorous proof but rather as a motivation. For instance, we describe a two dimensional discretized connected and compact surface as a simplex of equilateral triangles, where at each vertex is attached a delta function of curvature depending on the number of triangles meeting at the vertex. The conical singularity is proportional to the deficit angle $\Delta = 2\pi - n\frac{\pi}{3}$. We may assume the following correspondence

$$\int Dg_{\mu\nu}^{(h)} \rightarrow \sum_{\Lambda_h}, \quad (2.1)$$

and then the integral over the matter field X in an sum over all the embedding on the real line at each vertex. Finally, denoting a the size of the sides of the triangles, we may assume the following dictionary

$$\begin{aligned} \int_{\mathcal{M}_h} d^2x \sqrt{g} g^{\mu\nu} \partial_\mu X \partial_\nu X &\rightarrow \sum_{\langle ij \rangle} \left(\frac{X_i - X_j}{a} \right)^2, \\ \int_{\mathcal{M}_h} d^2x \sqrt{g} \mu &\rightarrow \frac{\sqrt{3}}{4} a^2 \mu V. \end{aligned} \quad (2.2)$$

where V denotes the number of vertices. We call $\kappa = \exp\left(-\frac{\sqrt{3}}{4} a^2 \mu\right)$ The partition function can thus be written as

$$\mathcal{Z} = \sum_{h \geq 0} (g_0)^{-\chi_h} \sum_{\Lambda_h} \kappa^V \prod_{i=1}^V \int dX_i \prod_{\langle ij \rangle} G(|X_i - X_j|). \quad (2.3)$$

We introduced $G(|X_i - X_j|) = \exp \left[- \left(\frac{X_i - X_j}{\sqrt{4\pi a}} \right)^2 \right]$. We may define the continuum limit as the double scaled limit where $a \rightarrow 0$ and the number of vertices to infinity so that to keep the physical area fixed. At the level of the partition function this is obtained by taking $a \rightarrow 0$ and κ to a critical value κ_c where the infinite sum shows a singular behavior. Progress can be made in evaluating such a sum by identifying it with the Feynman diagram expansion of a certain matrix model as it will be described in the following.

The matrix model Here we start with the matrix quantum mechanics defined as a euclidean path-integral over $N \times N$ hermitian matrices M with cubic potential $U(M) = \text{Tr} \left[\frac{M^2}{2} + \frac{M^3}{3!} \right]$. We also suppose that the action is defined on the compact slice of euclidean time $[-T, T]$. All in all, the partition function reads

$$\mathcal{Z}(\alpha, \beta) = \int_{M(-\beta/2)=M(\beta/2)} DM(\tau) e^{-\alpha \int_{-\beta/2}^{\beta/2} d\tau \text{Tr} \left[\frac{M^2}{2} + U(M) \right]}. \quad (2.4)$$

Dictionary with the discretized sigma model By rescaling of the matrix we can rewrite the partition function as up to an irrelevant prefactor **PG: Is this really irrelevant because I found that the Jacobian makes appear a factor of $\tilde{\kappa}^N$?**

$$\mathcal{Z}(\alpha, \beta) = \int_{M(-\beta/2)=M(\beta/2)} DM(\tau) e^{-N \int_{-\beta/2}^{\beta/2} d\tau \text{Tr} \left[\frac{\dot{M}^2}{2} + \frac{M^2}{2} + \tilde{\kappa} \frac{M^3}{3!} \right]}. \quad (2.5)$$

Sending $\beta \rightarrow \infty$

Solving the matrix model Also, the partition function can be interpreted as a thermal partition of inverse temperature and thus we can write

$$\mathcal{Z} = \text{Tr}_{\mathcal{H}} \left[e^{-\beta \alpha H} \right]. \quad (2.6)$$

Here $\mathcal{H}$ denotes the Hilbert space of the theory and H corresponds to the hamiltonian. Setting the inverse temperature to infinity, the partition function reduces to the vacuum amplitude

$$\mathcal{Z} \underset{\beta \rightarrow \infty}{\sim} e^{-\beta \alpha E_{\text{vac}}}. \quad (2.7)$$

In particular, one finds

$$\frac{\log(\mathcal{Z})}{\beta} \underset{\beta \rightarrow \infty}{\rightarrow} -\alpha E_{\text{vac}}. \quad (2.8)$$

The objective is to compute exactly such quantity.

Now, we wish to use the symmetry under conjugation by a unitary matrix to diagonalize the matrix M . If we consider such symmetry as a gauge symmetry, this crucially alters the boundary condition we shall impose since configurations of M related by a unitary belong to the same gauge orbit. We postpone such subtlety for latter and consider such symmetry as a global symmetry. We diagonalize $M = \Omega_M \Lambda_M \Omega_M^\dagger$, where Λ_M is the diagonal matrix of eigenvalues and Ω_M is a unitary matrix **PG: What is the difference between using $SU(N)$ and $U(N)$? I guess you should add the identity as a generator?**. The measure is not invariant under such change of variables and one obtains

$$DM(\tau) = \prod_{I=1}^N D\lambda_I(\tau) D\Omega(t) \Delta^2(\lambda), \quad (2.9)$$

$$\text{Tr} \left[\left(\frac{dM}{d\tau} \right)^2 \right] = \text{Tr} [\dot{\Lambda}^2] + \text{Tr} [\Lambda, \dot{\Omega}_M \Omega_M^\dagger]^2.$$

Remark that $\frac{d\Omega_M}{d\tau}\Omega_M^\dagger$ is Lie-algebra valued. Let's suppose we use $SU(N)$ symmetry and not $U(N)$. We can thus decompose this Lie algebra quantity in terms of the generators of $SU(N)$. It has $N^2 - 1$ generators and the rank is $N - 1$. The Lie algebra consists in skew-hermitian traceless matrices. We take as generators of the Cartan diagonal imaginary matrices, and for the rest of the generators we consider antisymmetric real matrices $iT_{ij} = \delta_{ij} = -iT_{ji}$ and matrices such that $iS_{ij} = i\delta_{ij} = iS_{ji}$. Hence,

$$\dot{\Omega}_M \Omega_M^\dagger = \frac{i}{\sqrt{2}} \sum_{I < J} \dot{\alpha}_{IJ} T_{IJ} + \dot{\beta}_{IJ} S_{IJ} + \sum_{I=1}^{N-1} \dot{\alpha}_I H_I. \quad (2.10)$$

We thus find

$$\text{Tr} [\dot{M}^2] = \text{Tr} [\dot{A}^2] + \sum_{I < J} (\lambda_I - \lambda_J)^2 (\dot{\alpha}_{IJ}^2 + \dot{\beta}_{IJ}^2). \quad (2.11)$$

Discarding the volume of $SU(N)$, the partition function reads

$$\mathcal{Z} = \int_{\lambda_I(-\beta/2) = \lambda_I(\beta/2)} \prod_{I=1}^N d\lambda_I(\tau) \Delta^2(\lambda(\tau)) \exp \left[-\alpha \int_{-\beta/2}^{\beta/2} d\tau \sum_{I=1}^N \frac{\dot{\lambda}_I^2}{2} + U(\lambda_I) + \frac{1}{2} \sum_{I < J} (\lambda_I - \lambda_J)^2 (\dot{\alpha}_{ij}^2 + \dot{\beta}_{ij}^2) \right]. \quad (2.12)$$

Importantly, there is a remnant symmetry which is the permutation of the eigenvalues. The wavefunction should thus be symmetric under such permutation so we effectively describe bosonic particles. The important observation is that the system can be mapped to a system of N fermions. Remark first that the operator

$$\mathcal{O} = \sum_{I=1}^N \frac{\pi_I^2}{2} + U(\lambda_I) + \sum_{I < J} \frac{(\pi_{IJ}^2 + \tilde{\pi}_{IJ}^2)}{2(\lambda_I - \lambda_J)^2} \quad (2.13)$$

is not hermitian with respect to the non-flat measure. The π 's denote the momenta. But it is hermitian with respect to the flat measure. Nevertheless, the operator can be modified to produce a hermitian operator with respect to the non-flat measure. In fact, if we define $H = \Delta(\lambda)^{-1} \mathcal{O} \Delta(\lambda)$ then H is hermitian with respect to the non-flat measure i.e

$$\int \prod D\lambda_I \Delta^2(\lambda) \psi_1^* H \psi_2 = \int \prod D\lambda_I \Delta^2(\lambda) \psi_2 H \psi_1^*. \quad (2.14)$$

The eigenvalue problem can thus be written as

$$\left[\sum_I -\frac{1}{2\alpha^2 \Delta(\lambda)} \frac{\partial^2}{\partial \lambda_I^2} \Delta(\lambda) + U(\lambda_I) + \sum_{I < J} \frac{(\pi_{IJ}^2 + \tilde{\pi}_{IJ}^2)}{2(\lambda_I - \lambda_J)^2} \right] \psi(\lambda_1, \dots, \lambda_N) = E \psi(\lambda_1, \dots, \lambda_N). \quad (2.15)$$

Solving the eigenvalue problem yields an orthonormal basis with respect to the non-flat measure. Setting $\Psi = \psi \Delta(\lambda)$ produces a fermionic wavefunction. We can put in one to one correspondence to a theory of N free fermions. The lowest energy configurations do not interact as the interaction is manifestly positive. The eigenvalue problem for the fermions reads

$$\left[\sum_I -\frac{1}{2\alpha^2} \frac{\partial^2}{\partial \lambda_I^2} + U(\lambda_I) + \sum_{I < J} \frac{(\pi_{IJ}^2 + \tilde{\pi}_{IJ}^2)}{2(\lambda_I - \lambda_J)^2} \right] \Psi(\lambda_1, \dots, \lambda_N) = E \Psi(\lambda_1, \dots, \lambda_N). \quad (2.16)$$

The ground state is obtained by filling the N first lowest energy levels

$$E_{\text{vac}} = \sum_{n=1}^N \epsilon_n. \quad (2.17)$$

In the limit where the inverse planck coupling α goes to infinity, we can resort to a semi-classical approximation using Bohr-Sommerfeld quantization condition

$$\int_{\lambda_-}^{\lambda_+} d\lambda \sqrt{2(\epsilon_n - U(\lambda))} = \frac{\pi n}{\alpha}. \quad (2.18)$$

In other words, in the semi-classical limit, the number of fermions is equal to the area of the phase space filled by the fermions

$$n = \int \frac{dp d\lambda \alpha}{2\pi} \Theta(\epsilon_n - \epsilon(p, \lambda)), \quad (2.19)$$

with $\epsilon(p, \lambda) = \frac{p^2}{2} + U(\lambda)$. In particular, denoting the level of the fermi sea as μ_F , we obtain

$$\int_{\lambda_-}^{\lambda_+} d\lambda \sqrt{2(\mu_F - U(\lambda))} = \frac{\pi N}{\alpha} = \pi \kappa^2(\mu_F). \quad (2.20)$$

$\lambda_{\pm}$ are the turning points of the classical orbits. Clearly, if one tunes μ_F close to the quadratic local maximum of the potential, $\kappa(\mu_F)$ develops a non analytic behavior.

A Boundary states: Ishibashi states vs Cardy states

We first introduce Virasoro primary states. Highest-weight representations of the Virasoro algebra are generated from states $|V_P\rangle$ satisfying

$$L_0 |V_P\rangle = \Delta_P |V_P\rangle, \quad L_n |V_P\rangle = 0 \quad (n > 0), \quad \Delta_P = \frac{Q^2}{4} - P^2. \quad (A.1)$$

We do not impose any normalization condition on the states $|V_P\rangle$.

On a conformal representative of a manifold with boundary, preserving one copy of the conformal symmetry requires the absence of momentum flow through the boundary,

$$n_\mu T^{\mu\nu} = 0. \quad (A.2)$$

At the quantum level, this condition translates into the gluing condition

$$(L_n - \bar{L}_{-n}) |B\rangle = 0, \quad n \in \mathbb{Z}, \quad (A.3)$$

where the boundary state $|B\rangle$ is viewed as a closed-string state.

For each Virasoro representation generated by a primary state $|V_P\rangle$, there exists an Ishibashi state, denoted by $|P\rangle$, which solves the gluing condition. It can be written formally as

$$|P\rangle = \sum_{N \geq 0} \sum_{I, J \in \mathcal{P}_N} \left(S^{(N)} \right)_{IJ}^{-1} L_{-I} |V_P\rangle \otimes \overline{L_{-J} |V_P\rangle}, \quad (A.4)$$

where $\mathcal{P}_N$ denotes the set of partitions of the level N , $L_{-I} = L_{-i_1} \cdots L_{-i_k}$ for $I = (i_1, \dots, i_k)$ with $\sum_a i_a = N$, and $S^{(N)}$ is the Shapovalov matrix of Virasoro descendants at level N . In general, physical boundary states also called Cardy states need not to be Ishibashi states. Physical boundary states $|s_1\rangle, |s_2\rangle$ satisfy the open/close string duality Cardy condition. Let an annulus of unit width for which the circle has length β . Then

$$\begin{aligned} Z_{s_1 s_2}(\beta) &= \text{Tr}_{\mathcal{H}_{\text{open}}} [e^{-\beta H_{s_1 s_2}}] = \sum_h n_h^{s_1 s_2} \chi_h(\beta) \\ &= \langle s_1 | e^{-H/\beta} | s_2 \rangle. \end{aligned} \quad (A.5)$$

This condition should be understood as determining $n_h^{s_1 s_2}$.